

Estimating Potential-Dependent Physicochemical Properties at Metal–Electrolyte Interfaces Using Machine Learning Interatomic Potentials

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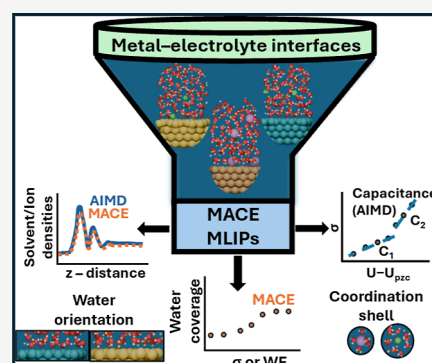
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ABSTRACT: Metal–electrolyte interfaces play a central role in electrocatalysis, energy storage, and environmental remediation. Understanding the structure and properties of these interfaces is therefore essential to designing efficient electrochemical systems. Density functional theory (DFT)-based molecular dynamics (MD) can accurately capture interfacial structure but is restricted to short time scales and small system sizes. To overcome these limitations, we develop machine learning interatomic potentials (MLIPs) using the MACE architecture within an active learning workflow to model aqueous NaCl electrolytes in contact with Au, Cu, and Rh(111) electrodes. The resulting committee of MLIPs achieves DFT-level accuracy across 21 metal–electrolyte systems spanning a wide range of surface charge densities. MACE–MD simulations reproduce key interfacial properties obtained from ab initio MD, including water density profiles, water orientation, and chemisorbed water coverage. Our simulations reveal a universal trend across all metals: the total coverage of water and ions decreases with increasing surface charge density or potential, reaches a minimum at or slightly below the *pzc*, and increases thereafter. Two distinct capacitance regimes emerge for all electrodes, corresponding to potentials below and above this point. Ion-specific effects strongly influence interfacial structure. Cl^- exhibits significantly stronger interactions with all metal surfaces than Na^+ , undergoing partial desolvation of up to 3.5 water molecules upon approaching the interface, compared to only 0.5 for Na^+ . These behaviors manifest in the vibrational density of states, where Cu and Rh show broad O–H stretching features at negative charge densities associated with Na^+ accumulation and strengthened hydrogen bonding. Overall, this work demonstrates that MLIPs based on the MACE architecture enable long-time scale, first-principles-accurate simulations of metal–electrolyte interfaces and provide detailed mechanistic insight into their potential-dependent physicochemical properties.

KEYWORDS: metal–electrolyte interactions, interfaces, machine learning interatomic potentials, water coverage, double layer capacitance, vibrational density of states



INTRODUCTION

A wide range of electrochemical processes, including solute adsorption,^{1,2} product desorption,³ proton–electron transfer,⁴ ion and solute diffusion,⁵ and double-layer formation,⁶ are controlled by the physicochemical characteristics of metal–electrolyte interfaces. Accurate descriptions of these interfacial properties are essential for guiding the design of next-generation systems for energy conversion, sustainable chemical and fuel production, and pollution remediation.^{7–9} For example, the molecular orientation and surface coverage of interfacial water during electrocatalysis strongly influence reaction rates,^{10,11} and in battery systems the structure of this interface plays a critical role in enabling or preventing dendrite growth.¹²

Extensive experimental studies have sought to elucidate the atomistic structure and dynamics of water at electrified interfaces. X-ray scattering measurements probe the organization and density of interfacial water layers,¹³ whereas surface-enhanced infrared absorption spectroscopy and Raman

spectroscopy provide detailed information on hydrogen bonding and molecular orientations near electrode surfaces.^{14,15} Nevertheless, achieving the spatial and temporal resolution required to characterize these properties in operando across a range of applied potentials and electrolyte environments remains difficult, hindering a comprehensive understanding of how interfacial behavior depends on potential and electrolyte composition.

Significant progress in computational methodologies has enhanced our understanding of metal–electrolyte interfaces. Density functional theory (DFT) simulations incorporating

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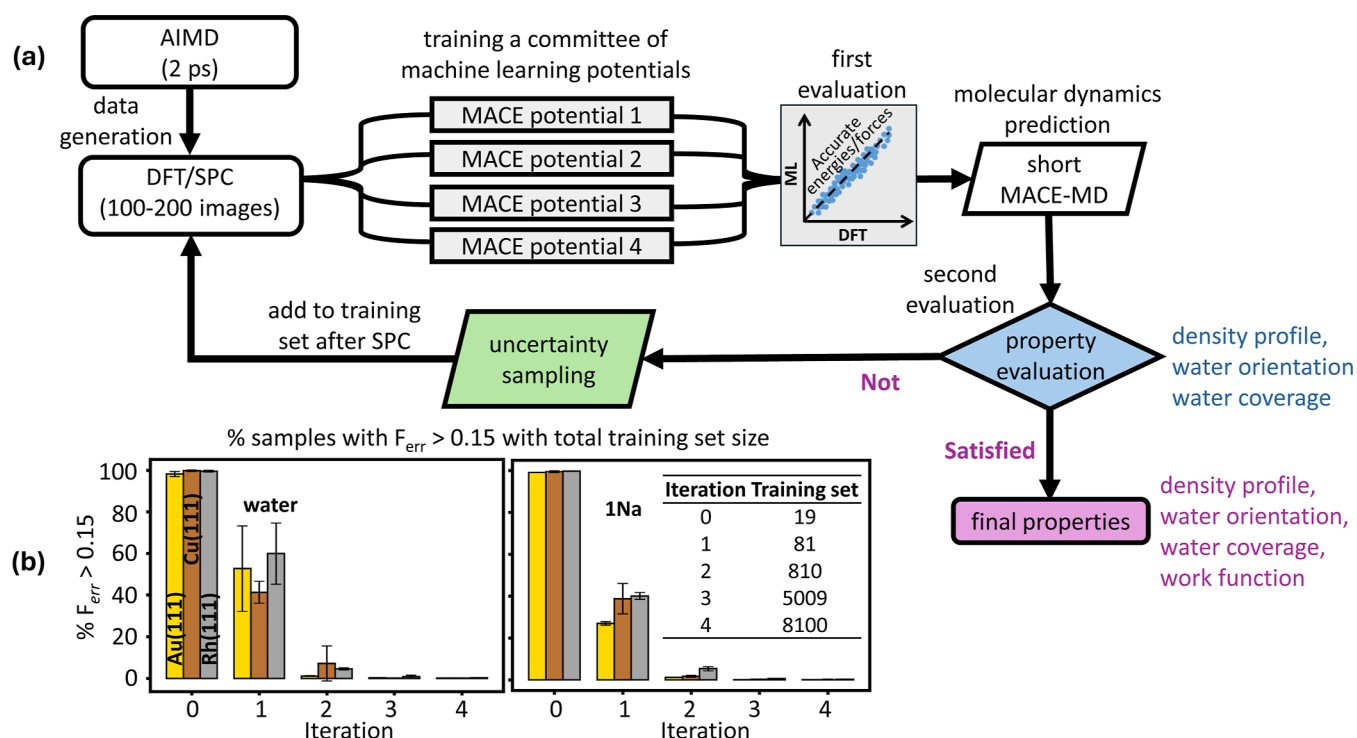


Figure 1. (a) Workflow for constructing the MLIP committee to simulate a metal-electrolyte interface. The active learning workflow consists of the following sequential steps: data generation, model training, first evaluation, molecular dynamics prediction, second evaluation, and uncertainty sampling. (b) Percentage of configurations from MACE-MD with uncertainty metric $F_{err} > 0.15$ against iteration with different numbers of training points (ran with MLIPs trained with training data from all 21 systems).

electrolyte effects, such as implicit solvation models,^{16,17} semi-implicit approaches that account for microsolvation,¹⁸ and quantum mechanics/molecular mechanics (QM/MM) techniques,^{19,20} have been widely applied to examine metal-electrolyte interfaces. Although implicit solvation models offer excellent computational efficiency and reproduce qualitative trends in solvation energetics, they cannot capture dynamic interfacial phenomena including solvent density fluctuations, molecular diffusion, and explicit solvent-solute interactions.^{21,22} QM/MM schemes provide quantum-level accuracy for explicit solvent molecules but require careful treatment of the boundaries separating the QM and MM regions.^{23,24} Ab initio molecular dynamics (AIMD) remains a powerful tool for resolving the explicit structure and dynamics of interfacial water; however, its high computational cost scales rapidly with system size, restricting most simulations to a few water layers and short trajectories of 10–100 ps.²⁵ Consequently, despite advances across these methods, achieving long-time scale and large-length-scale simulations of electrified interfaces with DFT-level accuracy remains an outstanding challenge.

Recent advances in machine learning methodologies^{26–29} have facilitated the development of machine learning interatomic potentials (MLIPs) which, when trained on DFT-derived energies and forces, accurately reproduce the potential energy surfaces of atomistic systems.^{28,30–37} These models enable accurate and computationally efficient simulations of electrolyte behavior and interfacial reactions at electrolyte–solid interfaces.^{38–40} MLIPs based on the MACE framework³⁴ have been applied to study the structural dynamics of aqueous electrolytes on Au(100) under varying surface charges and cation identities, achieving mean absolute errors of 0.25 meV atom^{−1} for energies and 8.5 meV Å^{−1} for

forces.³⁹ The MACE architecture has also been integrated with grand-canonical DFT to investigate the potential dependence of CO₂ reduction on Ag(111).⁴⁰

Understanding the potential-dependent structure of interfacial water is essential because aqueous metal–electrolyte interfaces play central roles in electrocatalysis and electrochemical devices. Copper and gold are state-of-the-art CO₂ reduction catalysts, with Cu uniquely capable of forming multi-carbon products and Au exhibiting high selectivity toward CO.^{41,42} Au is also a model system for the hydrogen evolution reaction.⁴³ Furthermore, Cu and Rh are important catalysts for bio-oil upgrading; Cu is active for electrocatalytic guaiacol hydrodeoxygenation and furfural reduction, while Rh effectively hydrogenates phenol.^{44–46} Although interfacial water structure on Au has been widely examined, comprehensive potential-dependent studies on other common catalysts such as Cu and Rh are more limited. Here, Cu, Rh, and Au are selected as representative 3d, 4d, and 5d transition metals to examine the potential-dependent structuring of water at metal–electrolyte interfaces.

A recent benchmark study of short-range MLIPs for charged metal–electrolyte interfaces reported that training on mixed-charge data can lead to observable-dependent errors, since the global surface charge is not a purely local quantity.⁴⁷ This study also found that equivariant, message-passing architectures such as MACE are considerably more robust than other ML models, and that the largest discrepancies arise for a subset of observables that are sensitive to charge, such as water orientation and dipole distributions. Our findings align with this general picture. We employ MACE and focus on local, near-surface observables that we explicitly benchmark against AIMD, including water density profiles, near-surface orientation distributions, and chemisorbed water coverage, while not

using MLIP-predicted work functions, as these are more strongly influenced by global electrostatics and require a more rigorous treatment of long-range electrostatic interactions.⁴⁸

In this work, we develop MLIPs using the MACE architecture³⁴ to perform long-time scale (2 ns) MD simulations with DFT-level accuracy to investigate Au, Cu, and Rh(111) metal–electrolyte interfaces and estimate the potential-dependent coverage of water and ions at these surfaces. We analyze multiple interfacial properties, including water and ion density profiles, water orientation distributions, ion coordination environments, and the vibrational density of states of interfacial water. Our results show that water coverage exhibits an S-shaped dependence on surface charge density, a feature absent in short-time scale AIMD simulations. The predicted ion density profiles reveal distinct near-surface distributions, demonstrating that Cl^- accumulates much closer to the electrode than Na^+ and that ions reside significantly closer to Cu(111) and Rh(111) surfaces than to Au(111).

■ COMPUTATIONAL METHODS

Seven distinct electrolyte compositions, with concentrations ranging from 0.5 to 2.5 M, were simulated in contact with Au(111), Cu(111), and Rh(111) electrodes, yielding a total of 21 systems. This concentration range aligns with recent studies investigating electrocatalytic interfaces and the promotional effects of ions.^{49–53}

To isolate potential-dependent interfacial structuring, we substituted interfacial water molecules with 1–2 Cl^- or 1–4 Na^+ ions (Table S1). We do not consider the coordinated interaction of Na^+ and Cl^- here; rather, they serve purely as model ions to probe ion-specific interactions (noting that chloride can specifically adsorb under some electrocatalytic conditions). The simulation cells maintained global charge neutrality without a homogeneous background charge. Here, σ is used as an effective ion-excess charge density descriptor, it does not represent an independently imposed excess electronic charge on the metal slab, but rather quantifies the net ionic charge introduced near the interface, which in turn modifies the interfacial electric field and molecular arrangement. Assuming all the counter charge is located on the surface atoms, we calculated σ by dividing the total counter charge by the total surface area. For the Au, Cu, and Rh systems, the resulting σ ranged from -35.03 to $17.52 \mu\text{C cm}^{-2}$, -45.78 to $22.89 \mu\text{C cm}^{-2}$, and -41.60 to $20.80 \mu\text{C cm}^{-2}$, respectively. To adequately sample distinct interfacial arrangements for each system, three independent initial configurations were generated by randomly removing water molecules and placing the ions at varying distances from the surface.

Active Learning Workflow

A committee of four MLIPs was developed to perform MD simulations. An active learning workflow, illustrated in Figure 1a, was employed to iteratively enrich the training dataset with diverse and informative configurations. Initially, the workflow was executed separately for each system, and a committee of MLIPs was trained on system-specific data. In the final stage, the aggregated dataset from all 21 systems was used to train a unified committee of MLIPs for production-scale simulations.

The active learning workflow consists of the following sequential steps: (1) data generation, (2) model training, (3) evaluation of energies and forces, (4) molecular dynamics prediction, (5) property evaluation, and (6) uncertainty sampling. Data generation relied on AIMD and DFT single-point calculations (SPCs) to create the starting training set. Next, we trained MLIPs using the MACE architecture (version 0.3.5), followed by the first evaluation stage to benchmark the model against DFT-predicted energies and forces. The MD prediction stage involved running short MD simulations using the trained MLIPs followed by a second evaluation stage for property evaluation by comparison against AIMD. The active learning loop concluded with an uncertainty sampling stage that identified configurations to be added to the training set.

To construct the initial dataset, three independent 2 ps AIMD simulations were performed for each system. Configurations were sampled every 20 fs and evaluated via DFT SPCs, resulting in approximately 100 initial training configurations. Four independent MACE models were then trained on this dataset, forming the MLIP committee. Performance was assessed using mean absolute errors (MAEs) in energies and forces from the MLIPs against SPCs-based test set, and a committee average was reported.

The trained MLIPs were subsequently used to conduct short MACE–MD simulations based on committee-averaged predictions for property evaluation and uncertainty-based sampling. Three independent MACE–MD simulations were initiated from the same starting configurations used for AIMD. For property evaluation, the first 10 ps of each trajectory (30 ps total) were compared with corresponding 30 ps AIMD trajectories. Evaluation focused on reproducing key AIMD descriptors, including (1) the peak positions of oxygen and hydrogen in interfacial water density profiles, (2) orientation distributions of first-layer water molecules, and (3) the coverage of chemisorbed water. Cosine similarity score and Pearson correlation coefficient were used to provide a quantitative comparison of the properties.

We used uncertainty-based sampling to augment the training set with informative MACE–MD configurations during each active learning iteration. Here, the uncertainty metric was based on the variance of the forces predicted by the MLIP committee, which represents the absolute disagreement among the models for the force per atom. A dimensionless relative uncertainty was defined by dividing the absolute uncertainty by the magnitude of the mean force (F_{err}) to select new configurations. During each iteration, 10–15% of the configurations with F_{err} values between 0.1 and 0.3 were selected for additional SPCs. Configurations with F_{err} below 0.1 were excluded as redundant, while those with F_{err} above 0.3 were omitted to avoid introducing structures that may fall outside the physically meaningful region of configuration space. As shown in Figure 1b, the proportion of configurations with high uncertainty decreased systematically with successive active learning iterations.

We stopped training MLIPs for each system once the property evaluations had converged, which required 400–450 training points per system. The combined data from all 21 systems was then used to train a single committee of four MLIPs for the MACE–MD production runs. We tested multiple MACE architectures with different numbers of interaction layers and hidden irreducible representations as a function of the total number of training points (Figure S1). Although the energy and force MAEs improved as more training points were added, the computational cost increased substantially. Because the different MACE architectures showed no significant differences in energy or force MAE, we used the ScaleShiftMACE model with two interaction layers and a hidden irreducible representation of “64x0e + 64x1o + 64x2e” for the production simulations. This model was trained on approximately 5010 configurations from the 21 systems, corresponding to about 238 configurations per system.

Density Functional Theory and Ab Initio Molecular Dynamics Simulations

Periodic DFT calculations were performed using the Vienna Ab initio Simulation Package (version 5.4.4)^{54,55} for SPCs and AIMD simulations. The projector-augmented wave method and a 450 eV plane wave kinetic energy cutoff were selected. The revised Perdew–Burke–Ernzerhof exchange correlation functional with Grimme D3 dispersion was used,^{56–58} which has been shown to reproduce the water behavior on metal fairly well.^{59–62} The energy convergence criterion for the electronic self-consistency cycle was 10^{-6} eV, and the maximum number of electronic iterations during a self-consistent field calculation was 250. A dipole correction perpendicular to the surface was included. We used the standard planar dipole correction scheme for slab geometries, and a fixed dipole reference point was specified at fraction coordinates (0.5, 0.5, 0.25) in the simulation cell. The lattice constants for bulk Au, Cu, and Rh were determined by relaxing a four-atom face-centered cubic cell using a k-point sampling grid⁶³ of $15 \times$

15×15 . The lattice constants for Au, Cu, and Rh were 4.19, 3.67, and 3.85 Å, respectively. The (111) facet of each metal was modeled using 24 surface atoms (6×4) with a thickness of four layers, where the bottom two layers were fixed during geometry optimization. The 6×4 surface cell, which is equal to or larger than most AIMD studies summarized in Table S8 and suitable for studying converged interfacial properties. The surface areas of the 6×4 Au(111), Cu(111), and Rh(111) slabs were $1.83 \times 10^{-18} \text{ m}^2$, $1.40 \times 10^{-18} \text{ m}^2$, and $1.54 \times 10^{-18} \text{ m}^2$, respectively. We used a vacuum layer of 30 Å above the electrolyte. The k -point sampling grid was $2 \times 3 \times 1$. The AIMD simulations for the initial training configurations and benchmarking data were generated using 1 fs time steps in the NVT ensemble using the Nosé-Hoover thermostat^{64,65} at 300 K.

MACE Architecture

We develop MLIPs using the MACE architecture (version 0.3.5), employing a ScaleShiftMACE model with two interaction layers and a hidden representation consisting of three irreducible components: 64 channels of even scalars (0e), 64 channels of odd vectors (1o), and 64 channels of even rank-2 tensors (2e). Local atomic environments were represented using the default MACE basis composed of eight spherical Bessel radial basis functions and real spherical harmonics up to $l = 2$, evaluated within a cutoff of 5 Å. The atomic energies for each isolated atom were calculated using least squares regression from the training data. The MACE models were trained using a loss function that combines energy and force errors, with weights of 1 and 100, respectively, to emphasize accurate forces for molecular dynamics stability. Each MLIP was trained for 1000 epochs using the Adam optimizer with an initial learning rate of 0.01. After 800 epochs, training entered a stochastic weight averaging phase in which the learning rate was reduced to 0.001 and the energy and force weights were set to 1000 and 100, respectively, to refine the energy landscape and improve the smoothness of the final model.

To quantify computational efficiency, we benchmarked a representative pure water–Au(111) system consisting of 96 water molecules on a 96-atom Au(111) slab (384 atoms total) using our trained MLIPs and compared the performance to AIMD. MACE-MD simulations performed on a single NVIDIA V100-32GB GPU (CUDA 12.4) with 4 CPU cores per task achieve approximately 0.86 ns/day using a single MLIP, while a committee of MLIPs achieves 0.34 ns/day due to multiple model evaluations per timestep. For reference, an AIMD simulation of the same system using VASP 5.4.4 executed on 12 compute nodes ((56 CPU cores per node, 672 total cores)) advanced only 0.8 ps/day. Under these benchmark conditions, the MLIP provides an effective speedup of approximately three orders of magnitude (single model) and over two orders of magnitude (committee) relative to AIMD.

Methods for MD Trajectory Analysis

Calculations of Oxygen, Hydrogen, and Ion Coverages.

Adsorption of a molecule or ion on an electrode in an electrolyte involves diffusion to the surface followed by displacement of chemisorbed water. Accurately determining the coverage of chemisorbed water is therefore essential for quantifying the enthalpic penalty associated with water displacement, which directly influences adsorption thermodynamics.^{2,66} Water may adsorb in either an oxygen-down or hydrogen-down configuration, corresponding to metal–oxygen and metal–hydrogen interactions, respectively. To quantify the chemisorbed water coverage, we evaluated the oxygen coverage (θ_{O_w}) and hydrogen coverage (θ_{H_w}), corresponding to the number of O_w and H_w atoms bound to the surface. These coverages were calculated by counting the number of O_w or H_w atoms within a characteristic chemisorption distance from the surface and dividing by the total number of surface atoms. Only chemisorbed atoms were included; physisorbed water molecules were excluded.

The characteristic chemisorption distances for oxygen on a metal ($r_{\text{O}_w, \text{M}}$) and hydrogen on a metal ($r_{\text{H}_w, \text{M}}$) were computed using eq 1⁶⁷ and eq 2, respectively. These distances depend on the covalent radius of the metal ($r_{\text{cov, M}}$) and of oxygen ($r_{\text{cov, O}}$) or hydrogen ($r_{\text{cov, H}}$). Values for $r_{\text{O}_w, \text{M}}$ and $r_{\text{H}_w, \text{M}}$ for each metal were defined relative to Cu (Table

1). Covalent radii for metals, oxygen, and hydrogen were taken from ref 68

$$r_{\text{O}_w, \text{M}} = 2.66 \text{ Å} \times \frac{r_{\text{cov, M}} + r_{\text{cov, O}}}{r_{\text{cov, Cu}} + r_{\text{cov, O}}} \quad (1)$$

$$r_{\text{H}_w, \text{M}} = 2.45 \text{ Å} \times \frac{r_{\text{cov, M}} + r_{\text{cov, H}}}{r_{\text{cov, Cu}} + r_{\text{cov, H}}} \quad (2)$$

Table 1. Covalent Radii of Selected Atoms and Characteristic Distances for Oxygen and Hydrogen Chemisorption

metal	$r_{\text{cov, M}}$ (Å)	$r_{\text{O}_w, \text{M}}$ (Å)	$r_{\text{H}_w, \text{M}}$ (Å)
Au	1.36	2.60	2.51
Cu	1.32	2.55	2.45
Rh	1.42	2.68	2.60
O	0.66	–	–
H	0.31	–	–
Cl	1.02	–	–
Na	1.66	–	–

Ions can also adsorb to the electrode and hinder reactant adsorption. Ion coverage (θ_{ion}) was quantified by defining a characteristic ion–metal distance, $r_{\text{ion, M}}$. For Cl^- , these distances were selected based on the ion density results on each surface in Figure 4a–c, yielding $r_{\text{Cl}^-, \text{M}}$ values of 3.35 Å, 2.65 Å, and 2.50 Å for Au, Cu, and Rh, respectively. The characteristic distances for Na^+ were then calculated relative to $r_{\text{Cl}^-, \text{M}}$ using eq 3, giving 4.25 Å, 3.37 Å, and 3.16 Å for Au, Cu, and Rh, respectively.

$$r_{\text{ion, M}} = r_{\text{Cl}^-, \text{M}} \times \frac{r_{\text{cov, M}} + r_{\text{cov, ion}}}{r_{\text{cov, M}} + r_{\text{cov, Cl}}} \quad (3)$$

The coverages θ_{O_w} , θ_{H_w} , and θ_{ion} were calculated as the number of adsorbed species per surface atom. The surface was defined as the topmost atomic layer (24 atoms), with its mean z -coordinate serving as the reference plane. A species was considered chemisorbed if its vertical distance from this plane was less than the characteristic distance. Coverage values were averaged over all frames from three independent trajectories to ensure robust statistics.

Work Function, Potential, and Capacitance Calculations. To benchmark the accuracy of the MACE–MD predicted work function (WF) and dipole moment along the surface normal (μ), we compared these quantities with AIMD-derived values for both neutral and charged Au(111) systems. For each method, 500 configurations were uniformly sampled from two independent 10 ps trajectories (250 frames from each trajectory, sampled every 40 fs), and single-point DFT calculations were performed to evaluate WF and μ . Figure S10 reports the distributions of μ , WF, the Fermi energy (E_{Fermi}), and the vacuum energy (E_{vacuum}) obtained from AIMD and MACE–MD for MLIPs trained on either the entire dataset or on system-specific data, evaluated at $\sigma = 0$ and $\sigma = 8.76 \text{ } \mu\text{C}/\text{cm}^2$. The WF distributions (Figure S10b,f) exhibit trends consistent with the corresponding dipole moment distributions (Figure S10a,e). MACE–MD predictions from globally trained MLIPs consistently overestimate the WF at both charge states, while system-specific MLIPs help improve mean WF predictions but yield distributions that are significantly broader than those from AIMD, indicating reduced precision despite accurate central tendencies. These discrepancies in the WF result from inaccuracies in both E_{Fermi} and E_{vacuum} . In particular, configurations obtained from MACE–MD simulations systematically underestimate E_{Fermi} compared to those obtained from AIMD simulations. These observations highlight the limitations of MLIPs for predicting WF, as the models are trained solely on atomic positions, energies, and forces and do not explicitly incorporate dipole and charges.

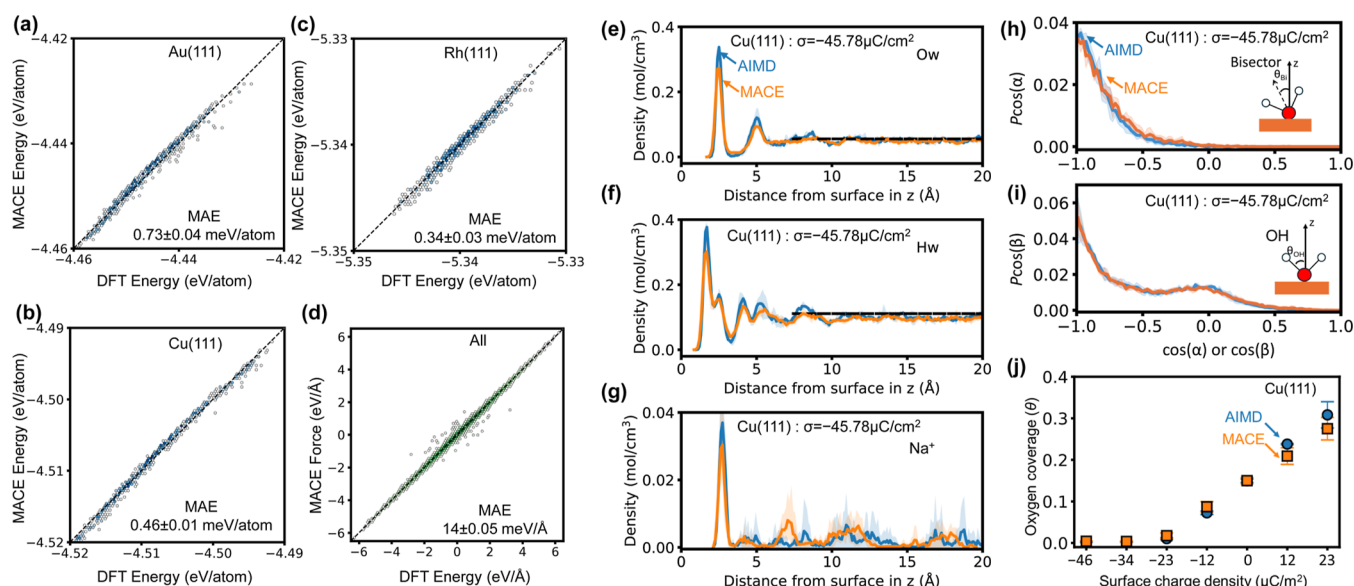


Figure 2. Benchmarking results for the MACE committee model. Parity plots of MACE-predicted energies for electrolyte systems containing (a) Au(111), (b) Cu(111), (c) Rh(111), and (d) MACE-predicted forces for all systems on a 2053 sample test set. Atomic density profiles of (e) oxygen and (f) hydrogen of the water molecules, and that of the (g) Na⁺ for the $-45.78 \mu\text{C}/\text{cm}^2$ electrolyte-Cu(111) system from AIMD and MACE-MD. Near-surface water orientational distributions for the electrolyte-Cu(111) system ($-45.78 \mu\text{C}/\text{cm}^2$), given as probability distributions of the cosine of the angle between the surface normal and (h) the water molecule bisector and (i) the O-H bonds of water. (j) Surface coverage of chemisorbed water on Cu(111) as a function of surface charge density. Each profile (e–j) is obtained from a 30 ps trajectory averaged over three independent 10 ps runs. The average density is represented by a solid line or point, while the statistical variance across the simulations is denoted by a shaded region or an error bar. The horizontal dashed line in (e–f) indicates the expected bulk number densities of solvent oxygen and hydrogen.

The deviations in MACE-MD predicted WF and μ are strongly linked to differences in the water-orientation distributions generated by each model. As shown in Figure S11, the predicted μ correlate directly with the net orientation of interfacial water molecules. This is further reflected in the orientation histograms shown in Figure S11c,d, where AIMD exhibits sharply peaked distributions, while MACE-MD produces broader distributions with systematic shifts. At $\sigma = 0$ and $\sigma = 8.76 \mu\text{C}/\text{cm}^2$, AIMD predicts μ values clustered near 0 eÅ and -0.5 eÅ, corresponding to peaks in the net water orientation ($\langle \cos(\alpha) \rangle$), near 0 and 10. In contrast, globally trained MACE models predict more positive μ accompanied by shifts of $\langle \cos(\alpha) \rangle$ toward more negative values, while system-specific models predict more negative μ associated with more positive $\langle \cos(\alpha) \rangle$. These systematic shifts demonstrate that inaccuracies in water-orientation sampling propagate directly into errors in the predicted μ and WF.

Due to inaccuracies in the MACE-MD predicted WF, we used only the configurations sampled from AIMD to evaluate the WF (and the corresponding potential vs pzc) of all the interfaces. We use the work function as an ensemble-averaged observable of interfacial electrostatics to establish relative trends across different surface charge densities rather than as a fully converged absolute electrode potential. The close agreement between the mean, median, and mode of the work-function distribution (Figure S12) confirms that the reported ensemble-averaged values are statistically stable and well-sampled. These WFs were used to determine the electrode potential, enabling estimation of the interfacial differential capacitance, which provides a quantitative measure of the potential-dependent charge-storage response at the metal–electrolyte interface. The potential for each system was computed from SPC-derived WF values applied to evenly sampled configurations from the AIMD trajectories, and was reported relative to the corresponding point of zero charge (pzc) using

$$U - U_{pzc} = \frac{WF - WF_{pzc}}{e} \quad (4)$$

The interfacial capacitance was calculated by performing a linear fit of surface charge density versus potential using the relation.

$$C = \frac{\sigma}{U - U_{pzc}} \quad (5)$$

RESULTS AND DISCUSSION

MACE MLIPs and MD Evaluation against AIMD

The performance of the MLIPs was benchmarked by comparing their energies, forces, and interfacial properties against DFT single-point calculations and AIMD trajectories. For each metal–electrolyte system, approximately 100 geometries were extracted from MACE-MD trajectories and evaluated with SPCs, yielding a combined test set of 2049 configurations. The MAEs in the energies for systems containing Au(111), Cu(111), and Rh(111) were 0.73 meV atom⁻¹ (Figure 2a), 0.46 meV atom⁻¹ (Figure 2b), and 0.34 meV atom⁻¹ (Figure 2c), respectively. The MAE for the energies and forces across all systems was 0.52 meV atom⁻¹ and 14 meV/Å (Figure 2d), respectively. The root mean square errors for energies and forces for all systems were 0.64 meV/atom and 23 meV/Å, respectively. These errors are consistent with recent MLIP studies of metal–electrolyte interfaces.^{39,50} System-specific MAEs are reported in Table S2. A breakdown of the per-element force errors is provided in Table S3, which are comparable to other benchmarking work for metal and liquid systems.^{30,34,69}

To further evaluate the ability of the MLIPs to capture interfacial properties, we compared the atomic density profiles of oxygen and hydrogen of the water molecules from MACE-MD with those obtained from AIMD. The orientational ordering and surface coverage of water molecules at the metal–water interface were also analyzed. For benchmarking, three independent 10 ps trajectories were used from both MACE-MD and AIMD simulations for each system. As a

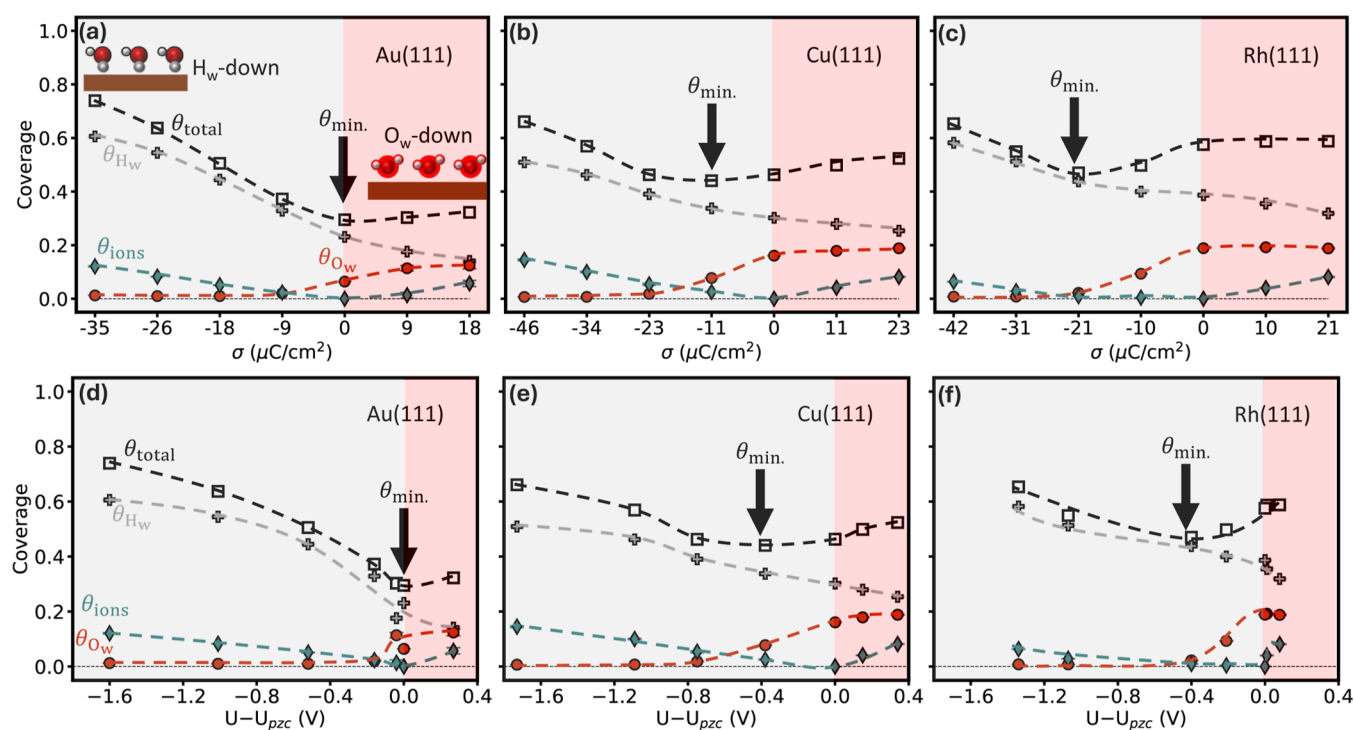


Figure 3. Surface coverage of water and ions on different metals. Coverage of oxygen-down (θ_{O_w}) and hydrogen-down (θ_{H_w}) oriented water, coverage of ions (θ_{ions}), and total coverage (θ_{total}) on (a) Au(111), (b) Cu(111), and (c) Rh(111) against surface charge density and on (d) Au(111), (e) Cu(111), and (f) Rh(111) against average surface potential relative to the pzc. Each scatter point represents a coverage for a given species, and the dashed lines connecting the points are drawn to guide the eye. The vertical arrow denotes the surface charge density or average surface potential that gives a minimum total coverage (θ_{min}). Error bars represent one standard deviation in coverage based on three independent 2 ns MACE–MD simulations sampled every 20 fs and are small on the scale of the plot.

sample test case, we plot the MACE–MD and AIMD predicted atomic density profiles (Figure 2e–g) and interfacial water orientation profiles (Figure 2h–i) for the electrolyte–Cu(111) system at a surface charge density of $-45.78 \mu\text{C}/\text{cm}^2$. We report the atomic densities and the interfacial water orientation comparison for each system in Figures S2–S9 of the Supporting Information.

We use the cosine similarity score ($\cos(\alpha)$) and Pearson correlation coefficient (r) to quantify the agreement between the properties predicted by MACE–MD and AIMD. The computed $\cos(\alpha)$ and r are summarized in Tables S5–S7 of the Supporting Information. The water coverage predicted by MACE–MD closely agrees with AIMD, with both $\cos(\alpha)$ and r near 1 for all systems. For atomic density profiles, the three highest peak positions in the average density were identified, and their $\cos(\alpha)$ and r values were compared. The average $\cos(\alpha)$ and r for both the oxygen and hydrogen density peaks across all systems ranged between 0.98–1 and 0.93–1, respectively. For the water orientation profiles, $\cos(\alpha)$ and r were evaluated over the full distributions of the angle bisector and O–H bond orientations of water. The average $\cos(\alpha)$ and r for the angle bisector and the O–H bonds of the near-surface water ranged between 0.98–0.99 and 0.9–0.98, respectively.

To obtain statistically meaningful interfacial properties, we employed the validated MLIPs to perform three independent 2 ns MACE–MD simulations (totaling 6 ns) for each metal–electrolyte system. This extended sampling enables ensemble averaging of local interfacial observables, including surface coverage and near-surface density profiles. As shown in Figure S13, short sampling windows (10–100 ps), comparable to typical AIMD time scales, can overestimate surface coverage.

In contrast, with extended sampling, these properties begin to converge after approximately 500 ps for all systems, providing improved statistical reliability of the reported trends.

Potential Dependent Coverage of Hydrogen, Oxygen, and Ions on Metals

The coverage of water and ions at the metal–electrolyte interface plays a crucial role in determining the local electric field, solvation environment and the adhesion of solution species to the surface, which in turn modulates adsorption energetics and reaction barriers.^{8,10} Chemisorbed water adopts either an oxygen-down or hydrogen-down configuration, corresponding to metal–oxygen and metal–hydrogen interactions, respectively. By calculating the coverage of oxygen, hydrogen, and Na^+/Cl^- (Figure 3), we obtained θ_{O_w} , θ_{H_w} , and θ_{ions} for all systems as functions of σ and U . The total coverage is defined as the sum of individual coverages: $\theta_{total} = \theta_{O_w} + \theta_{H_w} + \theta_{ions}$.

Beginning at negative σ and cathodic potentials, θ_{total} decreases as σ increases, reaches a minimum at or slightly below the pzc, and then rises again at more positive σ . The minimum coverage (θ_{min}), occurs at $\sigma = 0$ for Au(111) (Figure 3a) or $U = U_{pzc}$ (Figure 3d). For Cu(111) and Rh(111), θ_{min} shifts to negative σ or negative U . For all three metals, θ_{O_w} increases and θ_{H_w} decreases as σ or U becomes more positive. The θ_{O_w} values for Au(111), Cu(111), and Rh(111) range from 0.01–0.12, 0.01–0.19, and 0.01–0.19, respectively, whereas θ_{H_w} ranges from 0.14–0.61, 0.25–0.51, and 0.32–0.58. The point of inflection in θ_{O_w} as a function of σ is sharper

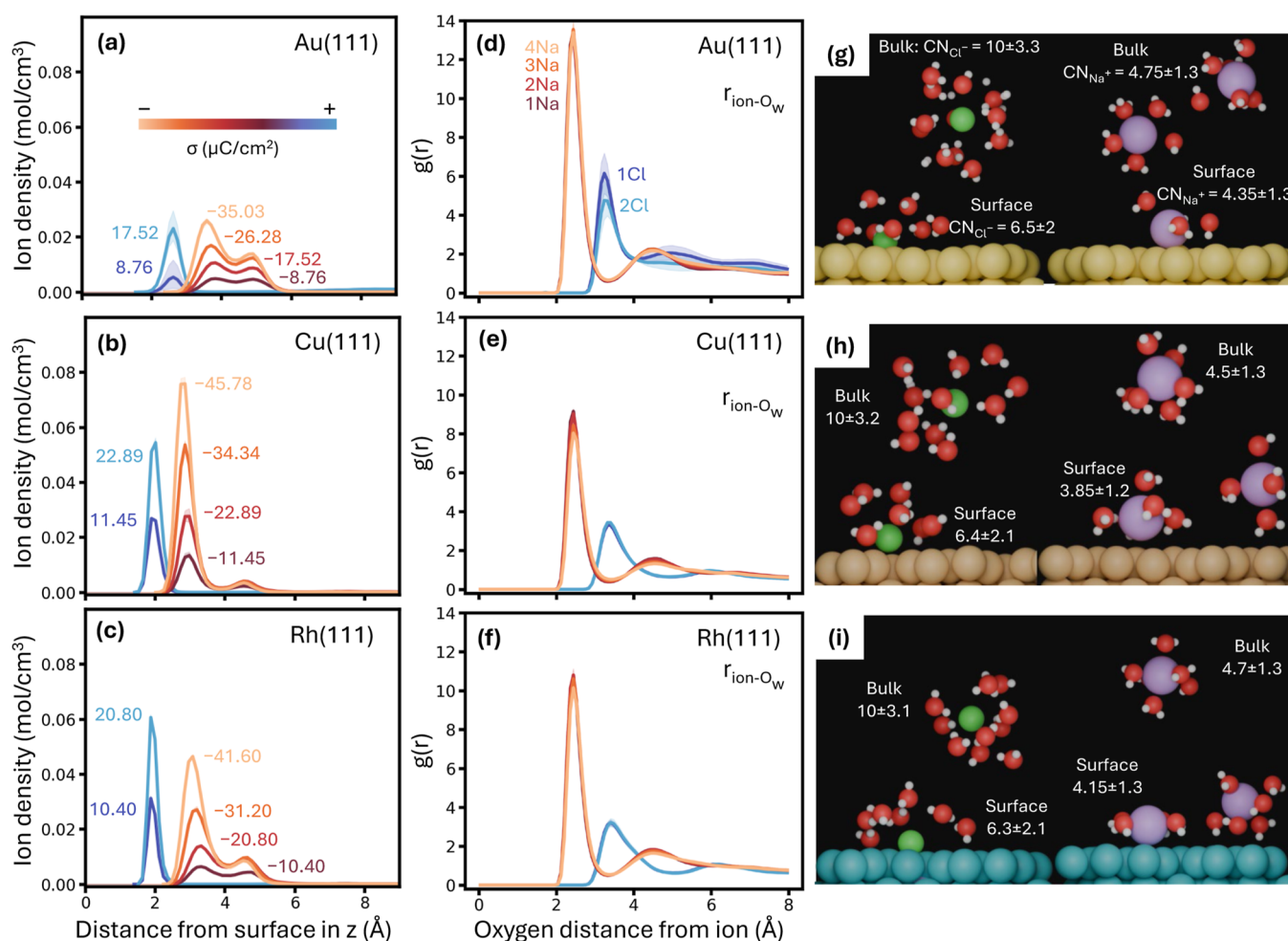


Figure 4. Ion atomic density profiles on (a) Au(111), (b) Cu(111), and (c) Rh(111) for different surface charge densities (ion concentrations). Radial distribution functions ($g(r)$) of water oxygen atoms around Na^+ and Cl^- in different ion concentrations on (d) Au(111), (e) Cu(111), and (f) Rh(111). Sample snapshots of Na^+ and Cl^- with their water coordination shell and values for near-surface and bulk coordination number on (g) Au(111), (h) Cu(111), and (i) Rh(111). The heatmap given in (a) shows the variation in surface charge density with ion type and number of ions. Each solid ion density and $g(r)$ profile is an average of three independent 2 ns MACE-MD simulations, and the shaded region denotes one standard deviation. The atomic radii used in the sample snapshots are for neutral sodium and chloride.

than that of θ_{H_w} , which decreases more gradually. The inflection in θ_{O_w} for Au(111) occurs close to $\sigma = 0$, where θ_{total} is equal to $\theta_{\text{min}} = 0.4$, whereas for Cu(111) and Rh(111), the inflection occurs at $\sigma < 0$ (Figure 3b,c), near the respective θ_{min} .

The θ_{total} at $\sigma > 0$ and the σ corresponding to the point of inflection in θ_{O_w} reflect intrinsic differences in metal–water interactions. The metal-dependent $\text{Rh} > \text{Cu} > \text{Au}$ trends observed at $\sigma > 0$, particularly under Cl^- -rich conditions, arise from a combination of intrinsic metal–water interactions and metal-dependent stabilization of specifically adsorbed ions. Even in the absence of explicit ions near the point of zero charge, Rh exhibits a stronger intrinsic interaction with interfacial water compared to Cu and Au, as reflected in its higher baseline water coverage. This intrinsic ordering is evident in the coverage analysis, where the point of inflection in θ_{O_w} as a function of σ occurs at progressively more negative surface charge densities moving from Au(111) to Cu(111) to Rh(111). For Au(111), the inflection appears close to $\sigma = 0$, coinciding with $\theta_{\text{total}} = \theta_{\text{min}} = 0.4$, whereas for Cu(111) and Rh(111) it shifts to $\sigma < 0$ (Figure 3b,c), near their respective

minimum coverages. The sharper inflection in θ_{O_w} relative to the more gradual decrease in θ_{H_w} highlights the dominant role of oxygen coordination in defining interfacial restructuring. At positive surface charge densities, this intrinsic metal–water interaction is further amplified by the preferential stabilization of Cl^- near the surface, leading to enhanced water accumulation and reinforcing the observed $\text{Rh} > \text{Cu} > \text{Au}$ trend.

Ion coverage behaves as expected: Na^+ dominates at $\sigma < 0$, and Cl^- dominates at $\sigma > 0$, with θ_{ions} maximized at both extremes. At $\sigma = 0$, we also evaluated systems containing one NaCl molecule on Au(111) and Cu(111) to assess its influence on water and total coverage. Compared to pure water, only small changes (0.01–0.06) were observed in θ_{O_w} , θ_{H_w} , and θ_{total} for both metals (Figure S14). Although the MLIPs were not explicitly trained on configurations containing Na^+ and Cl^- simultaneously, they perform well on dilute mixed-ion systems. Using 150 configurations sampled from a 1 ns trajectory, the energy and force MAEs were 1.62 meV/atom and 20.5 meV/Å for Au(111), and 0.85 meV/atom and 14.9 meV/Å for Cu(111).

For all metals, the minimum coverage of water and ions occurs at or slightly below U_{pzc} consistent with experimental and computational studies that report enhanced reactant adsorption near this potential.^{8,70} Bakshi and co-workers observed increased ethane coverage near U_{pzc} on Pt electrodes in 1 M H_2SO_4 , with adsorption decreasing at both more positive and more negative potentials.⁷¹ Similarly, Bockris and Jeng reported maximum coverage of various organic alcohols and acids between 0.3–0.5 V vs SHE on Pt.⁷⁰ Our MACE–MD simulations likewise predict that θ_{min} for the more noble Au(111) surface occurs at U_{pzc} , whereas for Cu(111) and Rh(111) it appears at $U - U_{pzc} = -0.4$ V. At more positive or negative potentials, θ_{total} increases due to enhanced adsorption of water and ions. The reduced coverage near U_{pzc} indicates a greater number of available surface sites for reactants, which may facilitate increased adsorption and catalytic activity in this potential region.

Interfacial Ion Accumulation and Solvation Shell Structure

Because the computational cost of DFT scales approximately cubically with system size, AIMD simulations are typically restricted to tens of picoseconds and simulation cells containing only a few hundred atoms.^{24,72} For interfacial electrolyte systems, this limitation generally permits the inclusion of only a small number of ions (typically 1–4, depending on the simulation cell volume and target concentration). Under such constraints, it is difficult to capture ion diffusion and to obtain statistically converged ion density profiles and coordination numbers as functions of surface charge density.^{17,73} In contrast, our system size of 92–96 water molecules and simulation time scale of 2 ns per trajectory is sufficient to capture ion specific properties and satisfies the criteria suggested in literature.⁷⁴ With three independent 2 ns MACE–MD simulations containing a comparable number of ions (1–4), we obtain well-converged ensemble-averaged ion properties and associated standard deviations. These trajectories enable the calculation of ion density profiles, ion– O_w and ion– H_w radial distribution functions ($g(r)$), and ion coordination numbers in both the near-surface region and the bulk.

The preferred distance of ions from the surface depends on the surface charge density (or surface potential) as well as the metal identity.^{39,75} In general, ions are located closer to Rh(111) and Cu(111) than to Au(111), consistent with stronger ion–surface interactions on the more reactive Rh and Cu surfaces. At positive surface charge density ($\sigma > 0$), the Cl^- density peak positions appear at 2 Å on Rh, followed by 2.14 Å on Cu and 2.55 Å on Au, reflecting the intrinsic metal–chloride interaction strength.

The preferred distance of ions from the surface depends on the surface charge density (or surface potential) as well as the metal identity.^{39,75} Across all metals studied, Cl^- consistently accumulates closer to the surface than Na^+ (Figure 4a–c). In general, ions are located closer to Rh(111) and Cu(111) than to Au(111), consistent with stronger ion–surface interactions on the more reactive Rh and Cu surfaces.

At positive surface charge density ($\sigma > 0$), the Cl^- density peak positions appear at 2.0 Å on Rh, followed by 2.14 Å on Cu and 2.55 Å on Au, reflecting the intrinsic metal–chloride interaction strength. Trajectory analysis (Figure S18a–c) further confirms the specific, stable adsorption of Cl^- at these close distances. This aligns with classical force-field

simulations that similarly suggest strong specific adsorption of Cl^- and negligible adsorption of Na^+ on Au(111).⁷⁶

Conversely, the weakly interacting Na^+ cations exhibit more distant (3–5 Å) and dynamically fluctuating behavior (Figure S18d–f). The Na^+ density on Au(111) is distributed between 3 and 6 Å, with the peak moving progressively closer to the surface as σ becomes more negative, consistent with the behavior of Na^+ and F^- on Ag(111).⁷⁵ On Cu(111), Na^+ densities are concentrated between 2 and 4 Å, with a peak near 3 Å and a tail extending to 6 Å, in agreement with Na–Cu distances reported from AIMD.⁷⁷ For all metals, the first Na^+ peak position shifts closer to the surface as σ becomes more negative. In contrast, short AIMD simulations with a single Na^+ on Cu(111) under applied potential report a mean ion distance that remains almost unchanged.⁷⁷

The radial distribution functions ($g(r)$) in Figure 4d–f show that O_w resides closer to Na^+ than to Cl^- . The $Na^+ - O_w g(r)$ has a first peak at 2.5 Å and a smaller secondary peak near 4.5 Å, whereas the $Cl^- - O_w g(r)$ exhibits a maximum near 3.5 Å. These peak positions are consistent with AIMD results reported by Bankura and co-workers.⁷⁸ Further evidence for the stronger water coordination around Na^+ is provided by the hydration-shell structures shown in Figure 4g–i, where Na^+ displays a compact and well-defined O_w coordination shell, while Cl^- exhibits a more diffuse solvation environment with H_w oriented inward.

Relative to the bulk, ion solvation decreases near the surface. We compute the average ion first-shell coordination number (CN_{ion}) in the bulk and at the interface by averaging over all σ values for a given metal. The bulk coordination numbers are independent of metal type. For all three metals, the bulk CN_{Cl^-} is approximately 10 ± 3 , decreasing by up to 3.5 water molecules as Cl^- approaches the surface and partially sheds its hydration shell. The relatively weak hydration of Cl^- is also reflected in the large standard deviation of CN_{Cl^-} . In contrast, the bulk CN_{Na^+} ranges from 4.50 to 4.75 and decreases by only 0.5–0.7 near the interface, which is consistent with the stronger water coordination around Na^+ compared to Cl^- .

Origin of Two Potential-Dependent Capacitance Regimes from Interfacial Water and Ion Rearrangement

The capacitance at metal–electrolyte interfaces varies sensitively with the applied potential.^{51,75,79–81} Potentials obtained from short AIMD simulations, followed by SPCs, are frequently plotted against surface charge density to estimate capacitance. Goldsmith and co-workers used AIMD with fixed surface charge and reported a capacitance of 14.3 $\mu F/cm^2$ on Au(111) over a potential range of –0.4 to 0.4 V relative to V_{pzc} .⁸² AIMD simulations including explicit Na^+ and Cl^- ions on Pt, Ag, and Cu predict either two distinct capacitive regions on a σ – U profile or a smooth, continuous potential dependence.^{51,75,83} Experimentally determined capacitances on Ag, Cu, Pt, and Rh likewise show potential-dependent profiles with a hump near the pzc and lower values away from this region.^{79,81,84,85} However, Garlyyev and colleagues observed the opposite trend for Pt and Au in perchlorate electrolytes, reporting a minimum near the pzc .⁸⁶ Here, we predict the capacitance of Au(111), Cu(111), and Rh(111) and interpret the trends in terms of water and ion rearrangements at the interface.

For all surfaces, two distinct σ versus U regimes emerge dynamically because of the metal-ion, metal-water, and water–ion interactions (Figure 5). Capacitances were extracted from

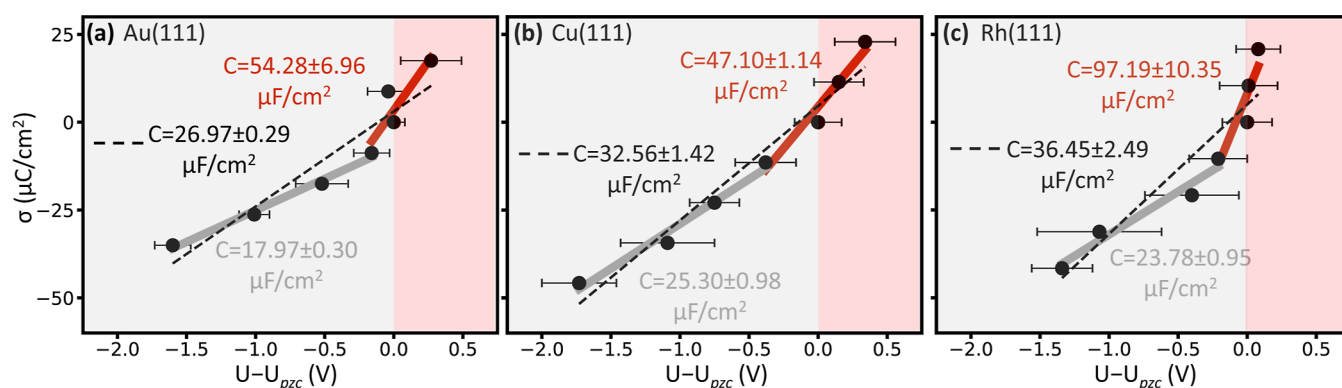


Figure 5. Surface charge density versus surface potential relative to the potential of zero charge on (a) Au(111), (b) Cu(111), and (c) Rh(111). The slope of the fitted line is capacitance ($C = \frac{d\sigma}{dU}$). We selected the capacitance fittings with the minimum average MAE for two regions after multiple trials. The solid lines represent a fit to a specific range of the data, whereas the dashed line is fitted to the entire dataset for comparison. Error bars denote one standard deviation of potential calculated from three independent 10 ps AIMD simulations sampled every 100 fs.

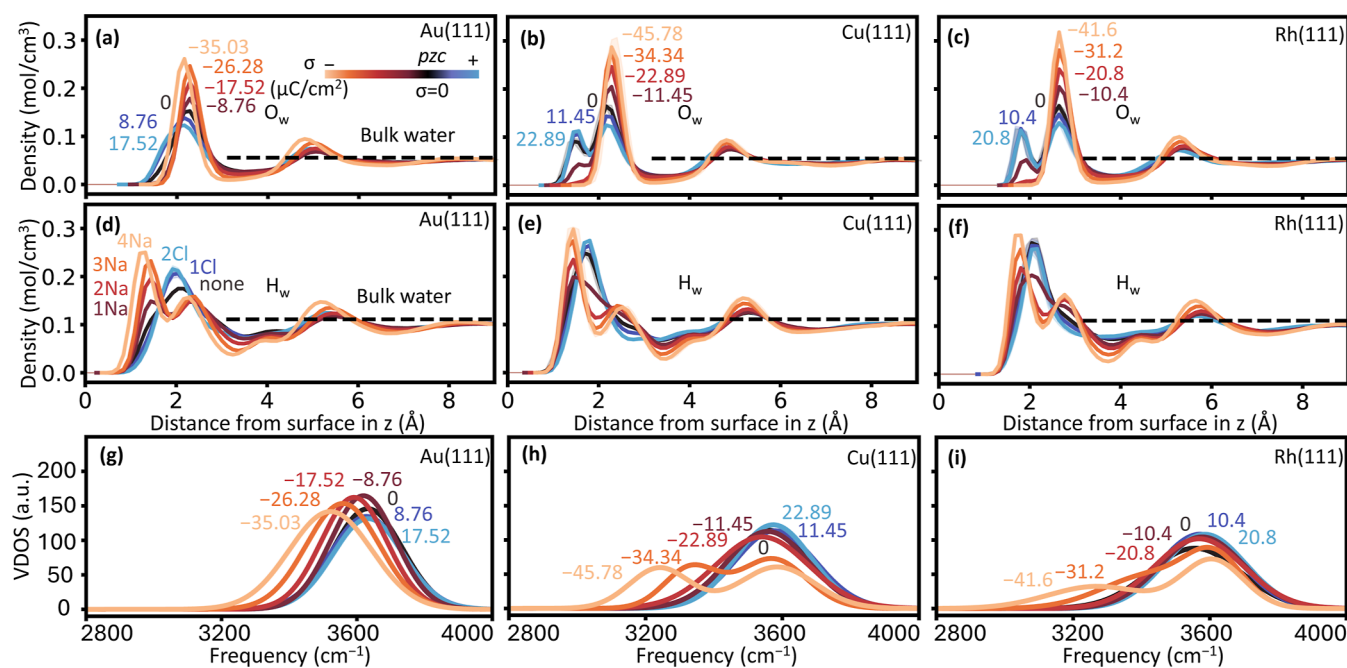


Figure 6. Atomic density profiles of (a–c) oxygen and (d–f) hydrogen of water molecules and on Au, Cu, and Rh(111) surfaces from MACE–MD. The O–H stretching modes from the vibrational density of states (VDOS) analysis of the near-surface water on (g) Au(111), (h) Cu(111), and (i) Rh(111). Each solid line represents an atomic density or VDOS for a specific ion concentration (surface charge density), and the heatmap given in (a) shows the variation in surface charge density with ion type and number of ions. Solid lines show the mean density profiles from three independent 2 ns MACE–MD simulations; shaded regions indicate the variation across replicates, although in many cases this variation is too small to be visually distinguished from the line thickness. The coordinate system is referenced such that $z = 0$ corresponds to the position of the uppermost metallic layer, and horizontal lines indicating the expected bulk densities of solvent oxygen and hydrogen are included for comparison. The dashed black line represents the bulk oxygen or hydrogen density for water. The VDOS analysis is conducted on a 40 ps MACE–MD simulation sampled every 1 fs and performed on the final configuration from a 2 ns simulation.

linear fits of σ versus U . We tried different linear fits of σ and $U - U_{pzc}$ and chose the pair of fits that yield minimum error. The capacitances of Au(111), Cu(111), and Rh(111) in low potential regions are 17.97 ± 0.30 , 25.30 ± 0.98 , and 23.78 ± 0.95 $\mu\text{F}/\text{cm}^2$, respectively. For $U \geq U_{pzc}$ these values increase substantially to 54.28 ± 6.96 , 47.10 ± 1.14 , and 97.19 ± 10.35 $\mu\text{F}/\text{cm}^2$. Two capacitance zones have also been reported in AIMD studies of Pt(111), with values near 100 $\mu\text{F}/\text{cm}^2$ slightly positive of the pzc , which decreases to approximately 20 $\mu\text{F}/\text{cm}^2$ at potentials farther from pzc .^{51,87} A similar trend has been reported for Ag.⁷⁵

A high capacitance for $U > U_{pzc}$ reflects a strong interfacial charge-storage response, where small changes in potential lead to large variations in surface charge. Conversely, at $U < U_{pzc}$, achieving a comparable change in surface charge requires a significantly larger potential perturbation. The two interfacial response regimes arise because of differences in ion interactions with the surface and a sharp increase in θ_{O_w} with σ (Figure S17). A high-capacitance regime at $U > U_{pzc}$ is therefore consistent with a more compact interfacial charge distribution arising from closer ion approach and enhanced interfacial water adsorption. In contrast, at $U < U_{pzc}$, weaker

water–surface and ion–surface interactions result in a reduced capacitance.

Near U_{pzc} , all three metals exhibit elevated capacitance, indicating a potential region where small changes in U yield large variations in σ . This behavior is related to the inflection in the θ_{O_w} profiles, where oxygen coverage rises sharply within a narrow potential window (Figure 3d–f). Because the surface charge is especially sensitive to the applied potential near the pzc , careful experimental control is required in this region. When capacitance is instead estimated using a single linear fit over the full potential range, we obtain values of 26.97 ± 0.29 , 32.56 ± 1.42 , and $36.45 \pm 2.49 \mu\text{F}/\text{cm}^2$ for Au(111), Cu(111), and Rh(111), respectively. These global fits slightly overestimate capacitances in the $U < U_{\text{pzc}}$ region and substantially underestimate them for $U > U_{\text{pzc}}$.

Interfacial Solvent Properties on Different Metals

Potential-Dependent Solvent Structure and Orientation at the Interface. To further elucidate the molecular origins of the coverage trends, we examined the distribution and orientation of interfacial water as a function of surface charge density. Oxygen and hydrogen density profiles extracted from three independent 2 ns MACE–MD simulations (Figure 6a–f) reveal clear σ -dependent restructuring of interfacial water. As the surface becomes more positively charged, the O_w density systematically shifts toward the metal surface, consistent with prior AIMD studies on Ag(111) and Pt(111).^{51,75,88} Across all metals, the first two water layers yield peaks between 1.5–3 Å and 4–6 Å. On Cu(111) and Rh(111), a third peak emerges below 2 Å for $\sigma \geq U_{\text{pzc}}$, reflecting enhanced O-down chemisorption on more reactive metal surfaces and explaining the rise in θ_{O_w} for these metals (Figure 3). The complete density profiles shown in Figure S19a–f exhibit pronounced layering within 5 Å of the metal surface, a statistically flat plateau between approximately 7–15 Å, and a smooth decay associated with the water–vacuum boundary beyond 17 Å. Importantly, in the intermediate region, the density remains bulk-like and is independent of σ .

The H_w density profile shows the opposite trend compared to the O_w density profile as shown in Figure 6d–f. At strongly negative charge densities ($\sigma < -8 \mu\text{C}/\text{cm}^2$), two H_w peaks appear within 3.5 Å: a primary peak near 1–2 Å associated with hydrogen-down binding and a secondary peak between 2–4 Å associated with in-plane hydrogen orientations. Similar signatures have been reported in AIMD simulations on Au(111) and Pt(111).^{14,51} As σ approaches and exceeds the pzc , these features collapse into a broad single peak, indicating progressive reorientation toward O-down configurations. Both oxygen and hydrogen densities decay to bulk-like values beyond approximately 8 Å.

To quantitatively characterize these orientational changes, we evaluated the angular distributions of water ($\cos(\alpha)$ for the bisector vector and $\cos(\beta)$ for each O–H bond) for water molecules in the first layer, second layer, and chemisorbed layer (Figures S20–S22). At negative σ , $P(\cos(\alpha))$ and $P(\cos(\beta))$ peak at $\cos(\alpha) = -0.86$ and $\cos(\beta) = -1$, corresponding to 150° and 180° , and indicating hydrogen-down orientations consistent with the H_w density analysis. At $\sigma = 0$, water orientations are broadly distributed without a dominant orientation, resulting in a disordered first-layer structure. At $\sigma \geq U_{\text{pzc}}$, chemisorbed water exhibits distinct peaks at $\cos(\alpha) = 0.5$ and $\cos(\beta) = 0.25$, corresponding to O-down configurations. These features are particularly pro-

nounced on Cu and Rh, reflecting their stronger water–metal interactions and the appearance of a third O_w density peak at short distances. The overall reorientation trends are consistent with experimental and computational studies showing potential-dependent water ordering on metal surfaces such as Ag and Pt.^{51,88,89}

Taken together, density and orientational analyses demonstrate that interfacial water undergoes a charge-dependent structural reorganization that is directly correlated with the potential-dependent behavior of θ_{O_w} and θ_{H_w} . These findings highlight the critical role of water reorientation in shaping the electrostatic and chemical environment at the metal–electrolyte interface.

Vibrational Density of States for Near-Surface Water.

Vibrational density of states (VDOS) analysis was performed to provide insight into the O–H stretching and H–O–H bending modes of interfacial water on different metals. In Figure 6g–i, we focus on the O–H stretching region, while the full VDOS results, including H–O–H bending and libration modes, are presented in Figure S24. We selected a 40 ps analysis window based on VDOS convergence tests (Figure S23); extending the window beyond this up to 500 ps does not alter peak positions, while substantially increasing data volume and input/output analysis cost as the frames now have to be saved every 1 fs for VDOS analysis (e.g., 40 GB for 500 ps at 1 fs frame rate).

Across all metals, the peak position of the O–H stretching band decreases with decreasing σ , in agreement with experimental observations for Au(111), Pd(111), and Pt nanoparticles.^{90,91} For systems with $\sigma \geq \text{pzc}$, the O–H stretching band narrows significantly and exhibits a dominant peak near 3600 cm^{-1} , characteristic of dangling O–H bonds at the interface that participate weakly in the hydrogen-bonding network.¹⁴ The peak positions are approximately 3650 cm^{-1} for Au, 3600 cm^{-1} for Cu, and 3575 cm^{-1} for Rh. Interactions between water and metal surfaces are known to induce red shifts in O–H stretching frequencies.⁹¹ The larger red shift observed for Cu and Rh compared to Au is attributed to stronger water–metal interactions on these more reactive surfaces, which weaken the O–H bonds of interfacial water.

In general, the O–H stretching peak for Au is narrower than that for Cu and Rh at positive potentials. This difference can be rationalized by the distinct interfacial solvent structures on these metals. On Au, Cl^- accumulates near 2.5 Å from the surface, whereas on Cu and Rh, Cl^- resides closer to the surface, near 2.0 Å. Additionally, the O_w density on Au exhibits one primary interfacial peak, while Cu and Rh show broader distributions with two peaks, including a prominent peak at shorter distances associated with O-down adsorption. For $\sigma \geq \text{pzc}$, the rearrangement of water and the proximity of Cl^- to the surface therefore produce a broad distribution of O–H stretching frequencies on Cu and Rh, while Au displays a single, narrower feature.

In the $3200\text{--}3600 \text{ cm}^{-1}$ region, characteristic of water with strong hydrogen bonding, we observe broad O–H stretching features for systems with $\sigma \leq \text{pzc}$.⁹² As σ becomes more negative, the O–H stretching distribution broadens and the peak position shifts to lower frequencies at rates between 48 and $79 \text{ cm}^{-1} \text{ V}^{-1}$ (Figure S25), consistent with the vibrational Stark effect.⁹³ Enhanced hydrogen bonding at potentials more negative than the pzc typically results in such red shifts.⁹¹

On Au(111), the O–H stretching region remains as a single broad band under all conditions, with its peak position shifting to lower frequencies as σ decreases. In contrast, Cu(111) and Rh(111) exhibit significantly broader distributions and multiple peak features at more negative σ . These broad O–H stretching regions, often with two distinct peaks, are linked to the presence of Na⁺ ions near the surface (Figure 4b,c) and to the reorientation of interfacial water into H-down configurations, consistent with our density and orientational analyses. The lower-frequency components can be attributed to trihedrally and tetrahedrally coordinated water structures at the interface, formed through a strengthened hydrogen-bond network under these conditions.⁹⁰

CONCLUSIONS

Long-time scale molecular dynamics simulations driven by machine-learning interatomic potentials enable statistically converged, ensemble-level predictions of electrolyte–metal interfacial properties at near DFT accuracy. Using MACE–MD, we characterized the structure, ion behavior, and vibrational signatures of water on Au(111), Cu(111), and Rh(111) across a wide range of surface charge densities and potentials.

For all metals, the total coverage of interfacial water and ions reaches a at or slightly below U_{pzc} . This reduction in coverage increases the availability of surface sites, consistent with prior reports of enhanced reactant adsorption near the pzc . Oxygen-down configurations become increasingly favored with increasing σ , resulting in an S-shaped increase in $\theta_{\text{O}_\downarrow}$.

Further, the capacitance profiles exhibit two regimes: a low-capacitance region for $U < U_{\text{pzc}}$ and a high-capacitance region for $U > U_{\text{pzc}}$. These regimes arise from potential-dependent restructuring of the interfacial layer. At positive potentials, Cl[−] accumulates within 2–2.5 Å of the surface and promotes dense packing of O-down water. Both Na⁺ and Cl[−] desolvate as they approach the interface, with Cl[−] losing up to 3.5 waters, leading to increased charge responsiveness. Near the pzc , small variations in U yield large changes in σ , highlighting the sensitivity of interfacial charge to subtle electrostatic perturbations.

Potential and ion composition also reshape the hydrogen-bond network. The O–H stretching band narrows near 3600 cm^{−1} at positive σ and broadens and red-shifts at negative σ , reflecting stronger hydrogen bonding and increased Na⁺ accumulation on Cu and Rh.

Overall, our results demonstrate that, after explicit benchmarking against AIMD, MACE–MD can accurately capture many local, near-surface, potential- and charge-dependent restructuring trends at electrolyte–metal interfaces, including water density profiles, chemisorbed water coverage, ion-specific accumulation trends, solvation-shell evolution, and vibrational signatures. However, accurate prediction of properties that depend directly on global electrostatics, such as water dipole distributions and work functions, remain challenging and might require the use of models with explicit long-range electrostatics.⁴⁸

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acselectrochem.5c00540>.

Supporting information contains system compositions and surface charge densities, learning rates for different MACE architectures, individual test dataset energy and force errors, benchmarking results for MACE–MD against AIMD, water and ion coverage on Au(111) and Cu(111) at pzc , RDF of water around Na⁺ and Cl[−], orientational distribution of water from MACE–MD, VDOS analysis of near surface water, method description of MD analysis, and sample pictures of all systems. All training and test datasets, molecular dynamics trajectories and sample scripts used in this work are available on the GitHub repository⁹⁴ (PDF)

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Author Contributions

The manuscript was written using contributions from all authors. All authors have approved the final version of the manuscript.

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Notes

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